

REDUCTION FORMULAS

Example 1

Find the reduction formula for $I_n = \int x^n e^x dx$

and hence evaluate $\int x^3 e^x dx$.

Solution:

This requires the method of integration by parts.

$$I_n = \int x^n e^x dx = x^n e^x - n \int x^{n-1} e^x dx$$

$$\begin{cases} \text{let } u = x^n, & v' = e^x \\ u' = nx^{n-1}, & v = e^x \end{cases}$$

$$I_{n-1}$$

$$\therefore I_n = x^n e^x - n I_{n-1}$$

For $n=3$,

$$I_3 = \int x^3 e^x dx = x^3 e^x - 3 I_2$$

$$= x^3 e^x - 3 [x^2 e^x - 2 I_1]$$

$$= x^3 e^x - 3 x^2 e^x + 6 [x e^x - I_0], \text{ but } I_0 = \int e^x dx = e^x$$

$$= x^3 e^x - 3 x^2 e^x + 6 x e^x - 6 e^x + C$$

Hence

$$\int x^3 e^x dx = e^x (x^3 - 3x^2 + 6x - 6) + C$$

Example 2Find the reduction formula for $I_n = \int \cos^n x \, dx$ hence determine $\int \cos^5 x \, dx$.Solution

$$I_n = \int \cos^n x \, dx = \int \underbrace{\cos^{n-1} x}_u \underbrace{\cos x}_{v'} \, dx = \sin x \cos^{n-1} x + (n-1) \int \cos^{n-2} x \underbrace{\sin^2 x}_{1-\cos^2 x} \, dx$$

$$\text{Let } u = \cos^{n-1} x, \quad v' = \cos x$$

$$u' = (n-1) \cos^{n-2} x (-\sin x), \quad v = \sin x$$

$$I_n = \sin x \cos^{n-1} x + (n-1) \int \cos^{n-2} x (1 - \sin^2 x) \, dx$$

$$I_n = \sin x \cos^{n-1} x + (n-1) I_{n-2} - (n-1) I_n$$

$$n I_n = I_n + (n-1) I_n = \sin x \cos^{n-1} x + (n-1) I_{n-2}$$

$$\therefore I_n = \frac{1}{n} \sin x \cos^{n-1} x + \frac{n-1}{n} I_{n-2}$$

for, $n=5$

$$I_5 = \int \cos^5 x \, dx = \frac{1}{5} \sin x \cos^4 x + \frac{4}{5} I_3$$

$$= \frac{1}{5} \sin x \cos^4 x + \frac{4}{5} \left[\frac{1}{3} \sin x \cos^2 x + \frac{2}{3} I_1 \right] \quad \begin{array}{l} \text{but} \\ I_1 = \int \cos x \, dx \\ = \sin x \end{array}$$

$$\therefore \int \cos^5 x \, dx = \frac{1}{5} \sin x \cos^4 x + \frac{4}{15} \sin x \cos^2 x + \frac{8}{15} \sin x + C$$

Example 3Find the reduction formula of $I_n = \int \sec^n x \, dx$ Solution

$$I_n = \int \sec^n x \, dx = \int \underbrace{\sec^{n-2} x}_u \underbrace{\sec^2 x}_{v'} \, dx = \tan x \sec^{n-2} x - (n-2) \int \sec^{n-2} x \underbrace{\tan^2 x}_{\tan^2 x = \sec^2 x - 1} \, dx$$

$$\text{let } u = \sec^{n-2} x \quad v' = \sec^2 x$$

$$u' = (n-2) \sec^{n-3} x (\sec x \tan x)$$

$$u' = (n-2) \sec^{n-2} x \tan x, \quad v = \tan x$$

$$I_n = \tan x \sec^{n-2} x - (n-2) \int \sec^{n-2} x (\sec^2 x - 1) \, dx$$

$$I_n = \tan x \sec^{n-2} x - (n-2) \underbrace{\int \sec^n x \, dx}_{I_n} + (n-2) \underbrace{\int \sec^{n-2} x \, dx}_{I_{n-2}}$$

$$(n-1)I_n = I_n + (n-2)I_n = \tan x \sec^{n-2} x + (n-2)I_{n-2}$$

$$\therefore I_n = \frac{1}{n-1} \tan x \sec^{n-2} x + \frac{n-2}{n-1} I_{n-2}$$

Hence

$$I_n = \int \sec^n x \, dx = \frac{1}{n-1} \tan x \sec^{n-2} x + \frac{n-2}{n-1} I_{n-2}$$
